2025_Hexaware_Actual_Test_7

Test Summary

- No. of Sections: 2
- No. of Questions: 90
- Total Duration: 90 min

Section 1 - Aptitude

Section Summary

- No. of Questions: 60
- Duration: 60 min

Additional Instructions:

None

Q1. Which of the following is the value of n if the number$2^{74}+2^{2058}+2^{2n}$ is a perfect square?

- 2010
- 2018
- 2012
- 2020

Q2. A boy wishes to cut as many pieces of rope of equal length as he can from pieces that are 104, 78 and 65 m long. If he wants the pieces to be as long as possible and does not like to waste any rope, then how long should each piece be?

- 13
- 9
- 7
- 3

Q3. If 10 men can do work in 6 days and 15 women can do the same in 5 days, Then 8 men and 5 women can together do the work in :

- 7 days
- 6 days
- 5 days
- 4 days

Q4. A, B, and C can together do a job in 9 days. C alone can do the job in 36 days. In how many days can A and B do 50% of the job working together?

- 12
- 6
- 15
- 20

Q5. A boat covers 40 km upstream and 90 km downstream in 5 hr. It can also cover 60 km upstream and 60 km downstream in 5 hr. The speed of the water current is

- 4 km/hr
- 5 km/hr
- 20 km/hr
- 25 km/hr

Q6. A train 700 m long is running at the speed of 72 km/hr. If it crosses a tunnel in 1 minute, then the length of the tunnel (in meters) is

- 700
- 100
- 500
- Data inadequate

Q7. A sum of rupees 4420 is to be divided between Rakesh and Prakash in such a way that after 5 years and 7 years respectively the amount they get is equal, if compounded annually. The rate of interest is 10 percent. Find the share of Rakesh and Prakash

- 2000, 2420
- 2420, 2000
- 2480, 2420
- 2210, 2210

Q8. If the rate increases by 2%, the simple interest received on a sum of money increases by Rs. 108. If the time period is increased by 2 years, the simple interest on the same sum increases by Rs. 180.The sum is :

Rs. 1800

Rs. 3600

Rs. 5400

Data inadequate

Q9. The cost of two varieties of paint is Rs.3,969 per 2 kg and Rs.1,369 per 2 kg respectively. After how many years will the value of both paint be the same, if variety1 depreciates at 26% per annum and variety2 appreciates at 26% per annum?

4

1

2

5

Q10. A man spends 35% of his income on food, 25% on children's education, and 80% of the remaining on house rent. What percent of his income he is left with?

8%

4%

5%

9%

Q11. Diwana buys a book from her friend Tarun, who sells the book by giving a discount of 20% from the price he fixed. But Tarun, being a cunning mind, increased the price of the book by 50% from the rate at which he bought, only from this price he gave a discount of 20 %. Diwana again sells the same book to her other friend for an extra 20Rs. Now finally this SP is actually 30% more than the original Cost price of the book. So, what is the profit of Diwana? (in percentage)

8.33%

6%

9%

10%

Q12. The ratio of the cost price and the marked price of an article is 14:17. If a 10% discount is given for the article, then find the profit percentage.

7%

9%

11%

13%

Q13. A boy running up a stair finds that when he goes up two steps at a time there is one step left over when he goes up three steps at a time there are 2 steps left over and when he goes up 4 steps at a time there are 3 steps left over. Find the number of stairs which is somewhere between 40 and 50.

41

43

47

48

Q14. A tin contains 48 liters of milk. Due to leakage, 720ml is lost. What percent of milk is still present in the tin?

98.6%

98.5%

97.5%

96%

Q15. A question paper has two parts P and Q, each containing 10 questions. If a student needs to choose 8 from part P and 4 from part Q, in how many ways can he do that?

None of the mentioned options

6020

1200

9450

Q16. A bag contains nine distinct yellow balls, three distinct white balls, and four distinct red balls. In how many ways can two balls be drawn from the bag?

$9C_2$

- $3C_2$
- $16C_2$
- $12C_2$

Q17. A letter is taken out at random from 'ASSISTANT' and another is taken out from 'STATISTICS'. The probability that they are the same letters is:

- 35/96
- 19/90
- 19/96
- none of these

Q18. In a simultaneous toss of two coins, find the probability of two heads.

- 1/4
- 1/2
- 3/4
- 4/5

Q19. The average age of a couple at the time of marriage was 24 years. After 9 years of marriage, they have a baby of 2 years old. Calculate the average age of the family when the baby was born.

- 16
- 15
- 19
- 22

Q20. The average monthly expenditure of Mr.Ravi's family for the first three months is Rs 2,750, for the next three months is Rs 2,940, and for the last three months Rs 3,150. If his family saves Rs 4980 for nine months, find the average monthly income of the family for the 9 months.

- Rs.3800

Rs. 3500

Rs. 3400

Rs. 4200

Rs. 4500

Q21. **Directions:**
What will come in place of question mark (?) in the following series?

6, 16, 34, 62, 102, ?.

Q22. **Directions:**
What will come in place of question mark (?) in the following series?

8, 12, 18, 27, 40.5, 60.75, ?.

Q23. Sailaja started from her house and walked 11 km towards the west followed by 20 km to her right. She then took a right turn to walk 6 km. Finally, she walked 8 km to her right. What is the distance between Sailaja's house and her destination?

17 km

12 km

15 km

13 km

Q24. P, Q, R, and S are playing a game of carrom. P, R and S, Q are partners. S is to the right of R who is facing West. Then Q is facing?

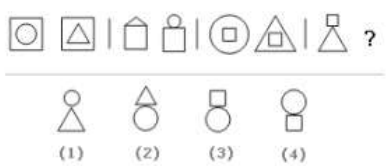
South

East

North

West

Q25. Observe the figures below and find out the one that comes in the place of "?".



1

2

3

4

Q26. **In the question below relationship between different elements is shown in the statements. The statements are followed by two conclusions.**

Statement: W>Y<X<Z=U>S; W<T ≥V

Conclusions:
I. Y<T
II. X > V

if only conclusion 1 is true

if only conclusion 2 is true

if either conclusion 1 or conclusion 2 is true

if neither conclusion 1 nor conclusion 2 is true

if both conclusion 1 and conclusion 2 are true

Q27. **In the question below relationship between different elements is shown in the statements. The statements are followed by two conclusions**.

Statement: X≤W=Y<P; W≥Z; Z ≥X

Conclusions:
I. P>X

II. Y≥X

if only conclusion 1 is true

if only conclusion 2 is true

if either conclusion 1 or conclusion 2 is true

if neither conclusion 1 nor conclusion 2 is true

if both conclusion 1 and conclusion 2 are true

Q28. In each of the questions below are given some statements followed by some Conclusions. You have to take the given statements to be true even, if they seem to be at variance from commonly known facts. Read all the conclusions and then decide which of the given conclusions logically follows from the given statements disregarding commonly known facts.

Statements:
Some liquid are solid
Some solid are gas
All gas are gasoline
Conclusions:
I. Some liquid are gasoline
II. All solid are not gasoline

- Only conclusion II follows
- Only conclusion I follows
- Both conclusions I and II follow
- Neither conclusion I nor conlclusion II follows

Q29. In the question below are given two or three statements followed by two conclusions numbered I and II. You have to take the given statements to be true even if they seem to be at variance from commonly known facts and then decide which of the given conclusions logically follows from the statements, disregarding commonly known facts. Give answer

Statements: Some gates are houses. All houses are ducks.
Conclusions:
I. All ducks are houses.
II. Some ducks are houses.

- If only conclusion I follows.
- If only conclusion II follows.
- If either I or II follows.
- If neither I nor II follows.

Q30. **Find the odd number out of the following number series.**

2, 4, 8, 11.5, 18.25, 28.375

- 8
- 11.5
- 18.25
- 28.375

Q31. **Find the odd man in the given series :**

61, 15, 53, 32, 27, 65

- 61

65

53

27

Q32. **Fill up the blank with suitable prepositions from the alternatives given under each sentence.**

Kanishka was initiated _____ Buddhism by Buddhist monks.

to

into

in

by

Q33. **Fill in the blank(s) with the most suitable option.**

If you do not comply _____ traffic regulations, you will get _____ trouble _____ the police.

about, into, for

with, into, with

over, in, with

on, to, for

Q34. **Fill in the blank with a suitable article from the options given.**

He has been appointed as _____ Minister for Industry.

a

an

the

no article

Q35. **Select the most suitable synonym**

Gratify

acknowledge

appreciate

indulge

appease

Q36. **Select the option that is most nearly OPPOSITE in meaning to the given word:**

FAR-FETCHED (OPPOSITE)

ingenious

facile

myopic

credible

Q37. Rajendra's royal _____ was proficient in creating icons of _____ style, working the hard granite to yield the vision of difficult themes.

avocation, picturesque

atelier, exquisite

fabricate ,crude

gallery, stratagem

Q38. **Fill in the appropriate adverb in the blank**

The gypsies do not live _____ at a particular place

willingly

permanently

voluntarily

none of the above

Q39. **A number of sentences are given below which, when properly sequenced, form a coherent paragraph. Each sentence is labeled with a letter. Choose the most logical order of sentences from among the four given choices to construct a coherent paragraph.**

A. Rather, as many view it, it would be inescapable.

B. With the number of flyers growing 6% annually and 3 billion cell phone devices in operation, foreign airlines are racing to offer passengers the ability to call, text, and e-mail with their mobiles and PDAs at about the same rates as for roaming international calls.
C. Connectivity has yet again despoiled once sacred ground, though this time it's in the air.
D. In the U.S., voice won't be available anytime soon, but Wi-fi will make e-mail and testing available next year.
E. Whether you're en route to London, Paris, or Dublin a fellow traveler's gritty life details, will soon be audible.

- CEABD
- BACDE
- EABDC
- BEACE

Q40. **In the questions below, each passage consists of six sentences. The first and sixth sentences are given in the beginning. The middle four sentences in each have been removed and jumbled up. These are labeled as P, Q, R, and S. Find out the proper order for the four sentences.**

S1: For some time in his youth Abraham Lincoln was manager for a shop.
P: Then a chance Customer would come.
Q: Young Lincoln's way of keeping shop was entirely unlike anyone else's
R: Lincoln would jump up and attend to his needs and then revert to his reading.
S: He used to lie full length on the counter of the shop eagerly reading a book.
S6: Never before had Lincoln had so much time for reading as had then.

- SRQP
- QSPR
- SQRP
- QPSR

Q41. **Read the sentence to find out whether there is any grammatical error in it. The error, if any, will be in one part of the sentence. The number of that part is the answer. If there is no error, the answer is (E), i.e. 'No error'.**

The brethrens (A)/ of this locality are always (B)/ at daggers dawn without any (C)/ apparent reason(D)/ No Error (E)

- (A)
- (B)
- (C)
- (D)
- (E)

Common Content:

Study the following information carefully and answer the question that follows:

J is the sister of T. T is married to D. D is the father of R.
M is the son of H. T is the mother-in-law of H.
D has only one son and no daughter.
J is married to B. L is the daughter of B.

Q42. How is J related to R?

- Sister
- Aunt
- Mother-in-law
- Mother
- Can't be determined

Q43. How is M related to T?

- Nephew
- Uncle
- Brother
- Son
- Grandson

Q44. How is L related to T?

- Niece
- Sister
- Can't be determined
- Mother

Common Content:

Each of the given question is followed by two statements, (A) and (B). Answer each question using the following instructions. Mark your answer as

1. If statement (A) alone is sufficient to solve the question, but statement (B) alone is not or If statement (B) alone is sufficient to solve the question, but statement (A) alone is not.
2. If neither statement (A) nor statement (B) is individually sufficient to solve the question, but a combination of both is sufficient to solve the question.
3. If both the statements (A) and (B) are individually sufficient to solve the question.
4. If both the statements taken together are not sufficient and more information is required to solve the question.

Q45. If k and n are integers, is n divisible by 7?
(A) $n - 3 = 2k$.

(B) 2k – 4 is divisible by 7.

1

2

3

4

Q46. What is the volume of cube X?
(A) The diagonal of one of the faces of X is $\sqrt{6}$.
(B) The diagonal of the cube from the upper rear left corner to the lower front right corner is 3.

1

2

3

4

Common Content:

Eight persons A, B, C, D, E, F, G, and H are sitting at the circular table all facing the center of the table. All the people are not necessary to be in the same order. A sits second to the left of B. Only two people sit between G and F, who is an immediate neighbor of B. More than two persons sit between H and D in both clockwise and anticlockwise directions. D is not an immediate neighbor of A. E is not an immediate neighbor of G. A does not face E. E does not sit second to the right of F.

Q47. Which of the following persons sits second to the left of H?

B

E

C

G

Q48. Who sits second to the left of the person facing B?

A

E

C

G

Q49. If H is related to D, B is related to E, than who among the following is related to G?

A

C

H

None of these

Q50. If all the people are made to sit in alphabetical order from clockwise direction of A, than how many people remains unchanged in their position except A?

One

Two

Three

None

Common Content:

Read the following passage and answer the questions that follow:

An eight-foot fibre-glass image of Goddess Durga and images of the allied pantheon, started their journey from Kolkata to Singapore last weekend for onward despatch to Dallas. The images should reach there by October, to be worshipped by the Indian community during the Durga Puja festival - the biggest event in the Bengali's social calendar.

Though there have been instances in the past of non-resident Indian communities organizing puja festivals abroad placing orders for images of the goddess from this city's artisans, it is not often that the chosen material for sculpting is fibre-glass. The relatively light medium makes transportation easy and inexpensive.

The artisans of Kolkata's Kumartuli, known for their dexterity in clay, have made the image. This is not the first time that artisan Prodyut Pal - member of a family that has been in the clay-modelling business for four generations - is responding to an overseas order. But it is his maiden attempt with fibre-glass.

Mr.Pal said: "Three years ago we sent a clay model of the goddess to Middlesex in the U.K. The organizers there have modified the practice of immersing the idol at the close of the festival, and it has been preserved in a home, to be brought out for community worship this year again - but for the last time. The order for a fresh image from the Puja committee in Middlesex for next year is expected.

Q51. The advantage of using fibre-glass for sculptures is that

it is light, making transportation easy and inexpensive.

it is easier than clay to work on.

it is environment-friendly.

it can be preserved for use the next year.

Q52. What is new about the Durga sculpture?

It is the first sculpture to go abroad.

It is made of fibre-glass.

It is the tallest image made.

For the first time, Durga Puja is to be celebrated outside India.

Q53. An order for a new image next year is expected from

Dallas.

Singapore.

Kumartuli.

Middlesex.

Q54. The organizers in Middlesex

do not celebrate Durga Puja.

ordered the fibre-glass image.

did not immerse the idol at the close of the festival.

use only clay-model images.

Q55. Mr.Prodyut Pal is

an artist in the clay-modelling business.

a manufacturer of fibre-glass.

an organizer of Puja in Kolkata.

a non-resident Indian.

Common Content:

Mobile banking (M banking) involves the use of a mobile phone or any other mobile device to _____1_____financial transaction linked to a client's account. M banking is new in most countries and most mobile payment models even in developed countries, to date operate on a _____2_____scale. A mobile network offers a _____3_____ available technology platform onto which other services can be provided at low cost with effective results. For example, M banking services that use _____4_____ such as SMS can be carried at a cost of less than one US cent per message. The low cost of using existing infrastructure makes such services more _____5_____ to use by customers with lower purchasing power and opens up access to services that did not reach them earlier due to the high cost of service delivery.

Q56. Choose the appropriate word for the blank numbered (3)

tangible

routinely

securely

unique

Q57. Choose the appropriate word for the blank numbered (1)

undertake

subscribe

lure

disburse

Q58. Choose the appropriate word for the blank numbered (2)

voluminous

rapid

limited

substantial

Q59. Choose the appropriate word for the blank numbered (4)

waves

process

connection

channels

Q60. Choose the appropriate word for the blank numbered (5)

amenable

valuable

answerable

exposed

Section 2 - Pseudocode and Computer Fundamentals

Section Summary

- No. of Questions: 30
- Duration: 30 min

Additional Instructions:

None

Q1. What will be the output of the following pseudocode?

```
1  Integer k,a,b,c,p
2  set k=8, a=1, b=2, c=3
3  p=k+b*c/k+b/c
4  if(p)
5     print a*b
6  else
7    print a/b
8  end if
```

- 4
- 3
- 2
- 0

Q2. What will be the output of the following pseudocode?

```
1  Integer a,b,c
2  Set a=4,b=4,c=4
3  for (each c from 2 to 4)
4     b=6+c
5     if((8-a+b)>(b+c))
6        b=(3+7)&b
7        a=(c+5)+b
8     Else
9        Jump out of the loop
10    End if
11    b=7+a
12 End for
13 Print a+b
```

- 31
- 20
- 24
- 27

Q3. What will be the output of the following pseudocode?

```
1  integer a, b, c
2  set a=1, b= 3, c= 2
3  if ((a&b) && (b&c) && (c&a))
4     a = a&b
5  else
6     b= b& a
7  end if
8  print a + b + c
```

8 print a+b+c

- 3
- 6
- 15
- 4

Q4. What will be the output of the following pseudocode?

```
1  Integer j, m
2  Set m=1
3  Integer a[4] = { 0,2,3,3}
4  if (a[0]+a[1]>a[2])
5     for (each j from 0 to 1)
6        if (a[j+1]>1)
7           m=1
8        Else
9           m=a[j+2]
10       End if
11    End for
12 End if
13 Print m
```

- 1
- 10
- -4
- 6

Q5. What will be the output of the following pseudocode?

```
1  Integer a,b,c
2  Set a=4,b=5,c=9
3  b=(c+6)+c
4  if((b^c^a)<(a^b))
5     b=(3+12)+c
6  End if
7  Print a+b+c
```

- 31
- 20
- 24
- 37

Q6. What will be the output of the following pseudocode?

```
Integer a,b,c
Set a=10, b=1, c=3
a=a+1
if (b mod c && c mod a)
   a=a+19
End if
Print a+c-b
```

- 5
- 32
- 2
- None of the mentioned options

Q7. What will be the output of the following pseudocode if a=5, b=2, and c=8?

```
Integer funn(Integer a, Integer b, Integer c)
   for (each c from 5 to 7)
      a=(b+b)^b
      a=9+b
   End for
   for (each c from 4 to 6)
      b=(a+b)^a
   End for
return a+b
```

- 67
- 54
- 57
- 58

Q8. What will be the output of the following pseudocode?

```
Integer a,b,c
Set a=4,b=2,c=15
a=a+b
b=10&b
b=(c&4)+a
Print a+b+c
```

- 31
- 28
- 40

45

Q9. What will be the output of the following pseudocode?

```
Integer a,b,c
Set a=3,b=4,c=4
for (each c from 4 to 8)
   a=(b^11)+b
   if((c-a)<(b-c))
      b=9^b
   Else
      Continue
   End if
   b=c+b
End for
Print a+b
```

18

12

74

70

Q10. What will be the output of the following pseudocode?

```
Integer p, q, r
set p= 1, q= 1, r= 2
q = p - 1
q = r - 1
if (p + 2 > q -6)
   if (q+2 > p- 6)
      p = 2
   end if
   q= 2
   if (r> 5)
      p = q
   end if
end if
printf p + q + r
```

3

6

10

17

Q11. What will be the output of the following pseudocode?

```
Integer a,b,c
Set a=9,b=9,c=8
c=c
if ((a+b)>(b+a)||4<a)
   b=(b+11)^c
   c=1
End if
Print a+b+c
```

51

19

46

38

Q12. What will be the output of the following pseudocode?

```
1  Integer a, b, c
2  Set a=1, b=4, c=6
3  c=(c+a)+a
4  a=(b+b)+c
5  c=(c^9)+a
6  a=11+a
7  Print a+b+c
```

48

51

54

44

Q13. What will the output of the following pseudocode be?

```
1  integer k, p
2  set k=24
3  p=k mod  5 + k / 10 * 10
4  if(p equals  k or p > 25)
5       if ( p equals 24 and  p <  23)
6            if (p equals 23)
7                 print p
8            end if
9            print p * p
10      enf if
11      print p * p * p
12 end if
```

It will print square of value of p

It will print square of value of p and cube of value of p

It will print cube of value of p

It will print value of p

Q14. What will be the output of the following pseudocode?

```
1  Integer a,b, c
2  Set a = 3, b = 4, c = 5
3  if ((b - c + a)< ( a - b ))
4    a = ( a + 9 ) & a
5    b = ( c & c) + a
```

```
6    if (( a + 6) >(c-a))
7     c = ( c +c) + c
8    End if
9    c = b + a
10   c = 8 + a
11  End if
12  Print a + b+c
```

- 8
- 12
- 22
- 16

Q15. What will be the output of the following pseudocode?

```
1   Integer m=0, n=13, flag=0, i
2   m = n/2
3   for(i = 2 to m)
4    if(n mod i=0)
5     flag = 1
6    go to line number 9
7    end if
8   end for loop
9   if(flag =0) // line 9
10   print n
11  else print 1
```

- 1
- 13
- 0
- 6

Q16. To implement all functions of the basic logic functions, it needs

- OR gate
- NOT gate
- AND and NOT gates
- None of the mentioned options

Q17. BCD numbers are obtained

- by converting decimal number to binary

- by converting decimal to octal
- when each decimal digit is represented by four bit binary
- by converting binary to decimal

Q18. The application layer of the OSI model is the

- seventh layer
- sixth layer
- fifth layer
- fourth layer

Q19. Which one of the following can produce the final product of machine processing in a form usable by humans?

- Storage
- Control
- Input device
- Output device

Q20. A data packet is a packet header together with

- A network layer
- An administrative layer
- User data
- A packet switch

Q21. A hexa decimal number 'A0' has the decimal value

- 80
- 256
- 100

160

Q22. The computer device primarily used to provide hardcopy is the

CRT

Line Printer

Computer Console

Card Reader

Q23. The term 'memory' applies to which one of the following?

Logic

Storage

Input device

Output device

Q24. The binary representation of hexadecimal 'C3' is

1111

110011

110001

11000011

Q25. How many units in a single bus structure will communicate at a time?

1

2

3

14

Q26. Which of the following is the most powerful type of computer?

- Main Frame
- Super Conductor
- Micro Computer
- Super Computer

Q27. Working with the WAN generally involves

- Satellite
- Frame Delay
- ATM
- User Agent

Q28. Which of the following component of a computer system is the most important to a database management system?

- Mouse
- High Resolution Video Display
- Printer
- High Speed, Large Capacity Disk

Q29. What device is used for entering x - y coordinates?

- Card reader
- Joystick
- Keyboard
- All of the mentioned options

Q30. What is the commonly used unit for measuring the speed of data transmission?

- Bits
- Band

Byte

None of the mentioned options

Answer Key & Solution

Section 1 - Aptitude

Q1

2020

Solution

2^74 + 2^2058 + 2^2n =(2^37)^2 + (2^1029)^2 + (2^n)^2

now, if we put 2^37 = a ; 2^1029 = b; Then for the above expression to be perfect

square 2^2n must be equal to (2*a*b)= 2*(2^37)*(2^1029);

==> 2^2n = 2^(1067)

Equating the powers:

==> 2n = 1067 ,

but this case is not possible since R.H.S is an odd integer whereas L.H.S is an

even integer.

So, the above mentioned case can't hold.

Now, if we put 2^37 = a; 2^n = b ; So, for the given expression to be perfect square

2^2058 = (2*a*b)= 2*(2^37)*(2^n) = 2^(n+38);

So, 2058 = (n+38)

=> n = 2020

So, The answer is n = 2020

Q2

13

Solution

To find the greatest possible common value among the given values, we have to find the HCF.

104 = 2 × 2 × 2 × 13

78 = 2 × 3 × 13

65 = 5 × 13

HCF of 104 m, 78 m and 65 m = 13 m

13 is the greatest length.

Answer - 13

Q3

5 days

Solution

10 Men can do the work in 6 days If the work is to be finished in 5 days, number of women required = 15

If the work is to be finished in 6 days, number of women required =[15×5]/6 = 25/2

So that 10 men = 25/2 women

or 5 women = 4 men 8 men + 5 women = 8 men + 4 men = 12 men

Now 10 men can finish the work in = 6 days

∴ 12 men can finish the work in [6×10]/12=5 days

Q4

6

Solution

In 9 days C does 9/36= 1/4 of work

So A & B do 3/4 of work in9 days

So A & B will take 4/3 x 9 = 12 days to do the complete job.

So 50% will take 6 days.

Q5

5 km/hr

Solution

Let, upstream speed = u km/hr

Downstream speed = d km/hr

40/u + 90/d = 5 (Time = Distance/Speed)

Similarly,

60/u + 60/d = 5

40x + 90y = 5 ...(i) (Assuming 1/u = x and 1/d = y)

60x + 60y = 5 ...(ii)

(Equation(ii) * 4) - (Equation (i)*6000), we get,

y = 1/30. So, x = 1/20

Hence, downstream speed = 30 km/hr

Upstream speed = 20 km/hr

So,

Speed of the boat in still water = 1/2*(30 + 20) = 25 km/hr

Speed of the current = 1/2*(30 - 20) = 5 km/hr

Q6

500

Solution

Let the' length of tunnel be x metres.

Speed of train = (72 × 5/18) m/sec = 20 m/sec

Time taken by the train to cover (700 + x) m = ([700 + x] / 20) sec.

[700 + x] / 20= 60

=> x = 500.

Q7

2420, 2000

Solution

Let the share of rakesh and prakash be R and P

$R*(1+10/100)^5 = (4420 – R)*(1+10/100)^7$

We get R = 2420, so P = 2000

Q8

Data inadequate

Solution

Rate of interest, Time and Simple interest is unknown in the first condition.

Rate of interest, time and interest is also not clear in the second condition too.

(Three unknown and two equation = solution cannot be determined)

Hence, data is inadequate.

Q9

2

Solution

We have cost of 2 kg of first variety = 3969.

Hence, cost of paint after 1 year = 3969-26% of 3969

= 3969 - 26/100 × 3969

= 3969 - 1031.94

= 2937.06

Cost of this variety after 2 years = 2937.06 – 26% of 2937.06

= 2937.06 – 26/100 × 2937.06

= 2173.4244

Cost of second variety = Rs 1369

Cost of paint after 1 year = 1369 + 26% of 1369

= 1369 + (26/100) × 1369

= 1369 + 355.94

= 1724.94

Cost of this variety after 2 years = 1724.94 + 26% of 1724.94

= 1724.94 + 26/100 × 1724.94

= 2173.4244

Hence cost of both the paints after 2 years will be same.

Q10

8%

Solution

Let the total income be x.

Then, income left = (100 - 80)% of [100 - (35 + 25)] % of x = 20% of 40% of x

= [(20/100) * (40/100) * 100] % of x = 8 % of x.

Q11

8.33%

Solution

Assume that cost price of book for Tarun = Rs. CTarun marks up the price of the book by 50%∴The marked price = C + 50% of C = 1.5CTarun allows a discount of 20% and sells it to Diwana.The amount Tarun receives after providing discount = marked price – 20% of marked price= 1.5C – 20% of 1.5C= 1.2C

∴ Diwana purchased it for Rs. 1.2C

Diwana sells it for Rs. 20 more than what she purchased for.The selling price of book for Tarun = Rs. (1.2C + 20)

This S.P is 30% more than the original C.P of the toy car.⇒ 1.2C + 20 = C + 30% of C⇒ 0.1C = 20⇒ C = Rs. 200

Now, Diwana's profit % is:

% Profit =[(1.2C + 20) - 1.2C] / 1.2C = 0.0833 * 100 = 8.33

Q12

9%

Solution

Let the CP and MP be 140X and 170X respectively.

SP after 10% discount = 170X * (9/10) = 153X

Required profit % = (153X - 140X) / 140X * 100 =9%

Q13

47

Solution

Had there been one more step, the number of steps would be an exact multiple of 2,3 as well as 4.

So LCM of 2,3,4 = 12

The highest common multiple in the range 40-50 will be 48

The number of steps is = 48-1 = 47

Q14

98.5%

Solution

Total oil in the tin = 48ltr

Oil leak out = 0.72ltr

Oil left in the tin = (48 - 0.72) = 47.28

Thus, Percentage of oil left in tin = (47.28/48) * 100 = 98.5%

Q15

9450

Solution

The students must choose 8 from part A and 4 from part B

10c8 x 10c4 = 45 x 210 = 9450

Q16

$16C_2$

Solution

Total number of balls = 9 + 3 + 4

Two balls can be drawn from 16 balls in $^{16}C_2$ ways.

Q17

19/90

Solution

ASSISTANT= AA I N SSS TT

STATISTICS =A II C SSS TTT

Here N and C are not common and same letters can be A, I,S, T. Therefore

Probability of choosing A =[2C1/9C1 x 1C1]/[10C1] = 1/45

Probability of choosing I = (1/9C1) x (2C1/10C1) = 1/45

Probability of choosing S = (3C1/9C1) x (3C1/10C1) = 1/45

Probability of choosing T = (2C1/9C1) x (3C1/10C1) = 1/15

Hence, required probability

= 1/45 + 1/45 +1/10 + 1/15 = 19/90

Q18

1/4

Solution

Sample space = (HH, HT, TH, TT)

Number of sample space n(S)= 4

There is only one chance to get two heads. that is HH

n(E) = 1

Probability = n(E) / n(S) = 1/4

Q19

22

Solution

Average age of couple is 24; Total age of the couple = 48 years

After 9 years of marriage they have a baby of 2 years old.

Hence, Total age of the family when the baby was born = (48 + 9 + 9 + 0) = 66

The average age of the family when baby was born is 66/3 = 22 years

Q20

Rs. 3500

Solution

Average monthly expenditure for 3 months = Rs. 2750

Total expenditure for 3 months = Rs 2750 x 3 = Rs. 8250

Average monthly expenditure for 3 months = Rs. 2940

Total expenditure for 3 months = Rs 2940 x 3 = Rs. 8820

Average monthly expenditure for 3 months = Rs. 3150

Total expenditure for 3 months = Rs 3150 x 3 = Rs. 9450

Total savings for 9 months = 4980

Average monthly income for 9 months = (8250 + 8820 + 9450 + 4980)/9 = 3500

Q21

156

Solution

Series Pattern Given Series

6 6

6 + 2 × 5 16

16 + 3 × 6 34

34 + 4 × 7 62

62 + 5 × 8 102

102 + 6 × 9 156

Q22 91.125

Solution

The given number series is based on the following pattern :

8×1.5=12

12×1.5=18

18×1.5=27

27×1.5=40.5

40.5×1.5=60.75

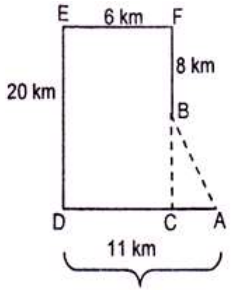
∴?=60.75×1.5=91.125

Q23 13 km

Solution

The path traversed by Sailaja is as fallows.

Let A be the starting point and B the destination



The distance required is AB = √(BC2 + AC2)

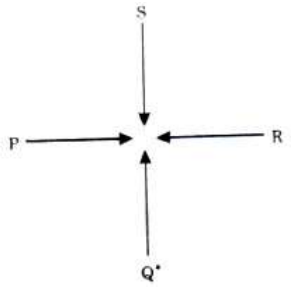
BC = CF – FB (or) DE – FB (or) 20 – 8 = 12 km

AC = AD – CD (or) AD – EF (or) 11 – 6 = 5 km

∴ AB= $\sqrt{(12^2 + 5^2)}$ = √(169) = 13 km

Q24 North

Solution



R is facing West, and S is to the right of R, so S is facing South.

Q is opposite to S so, Q is facing North

Q25

3

Solution

All four segments use the same figures: two squares, one circle, and one triangle. In the first segment, the squares are on the outside of the circle and triangle. In the second segment, the squares are below the other two. In the third segment, the squares on are the inside. In the fourth segment, the squares are above the triangle and circle. Correct answer is 3.

Q26

if only conclusion 1 is true

Solution

Combining inequalities V≤T>W>Y<X<Z=U>S

Clearly, T > Y. Thus, I is true

No relationship can be established between X and V.

Thus, II is not true

Q27

if both conclusion 1 and conclusion 2 are true

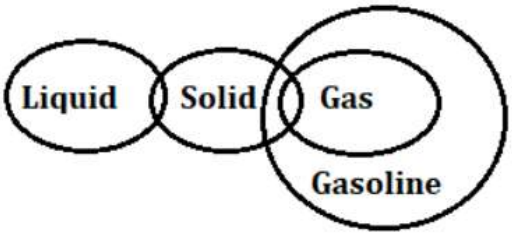
Solution

Combining the inequalities, X≤Z≤W=Y<P I and II both follow.

Q28

Neither conclusion I nor conlclusion II follows

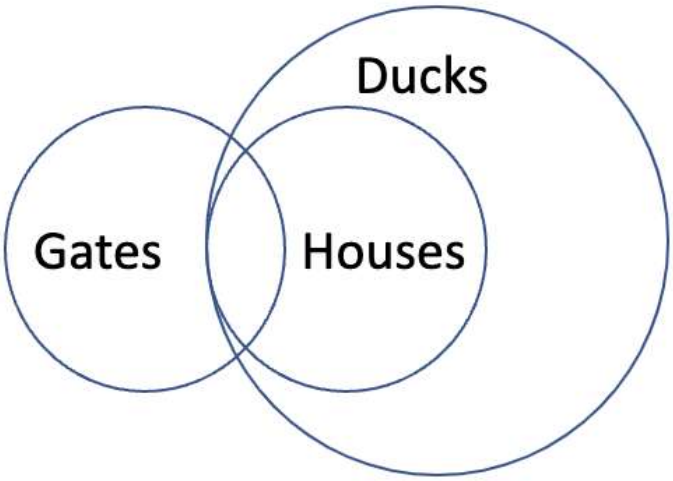
Solution



Q29

If only conclusion II follows.

Solution



Only II follows

Q30

8

Solution

Given number series follows a pattern that,

2

2 x 1.5 + 1 = 4

4 x 1.5 + 1 = 7 (not equal to 8)

7 x 1.5 + 1 = 11.5

11.5 x 1.5 + 1 = 18.25

18.25 x 1.5 + 1 = 28.375

Q31

65

Solution

Pattern for the series is :

1. All the numbers are reversed
2. Addition (or) subtraction is done alternately starting from 1.

i.e., 61 reverse is 16,

16 - 1 = 15

Then, 15 reverse is 51, so 51 + 2 = 53

Hence, 27 reverse is 72 to which 5 should be subtracted and we get 67

Hence, 65 is the odd man out

Q32

into

Solution

The preposition 'into' should be used after the word initiated.

Q33

with, into, with

Solution

Answer is obtained by looking at the first blank alone.The only preposition which can follow ‘comply’ is ‘with’.Comply with – Obey.

Q34

the

Solution

Before names of Professions such as minister, governor, president, prime minister, chief minister, etc. the definite article 'the' should be used.

Q35

indulge

Solution

The word gratify means obligate or please. Acknowledge refers to recognize, appreciate refers to glad about and appease refers to calm down. So the best answer is oblige which means obligate or please.

Q36

credible

Solution

Far-fetched means very unlikely to be true. Facile means simplistic. Myopic is lacking intellectual sight. Credible is acceptance. Ingenious means original.Ingenious is the antonym.

Q37

atelier, exquisite

Solution

The second blank must take a positive word, hence "Crude" and "stratagem" is not possible.

"Avocation" is hobby, hence that is not possible for blank 1.

Hence, Atelier means a workshop and exquisite means elegant . This is the best fit.

Q38

permanently

Solution

gypsies are nomadic people. Thus they do not permanently live in a place

Q39 CEABD

Solution

'C' is the opening sentence as it is a general statement introducing the topic of discussion - connectivity, with special reference to air. 'E' continues the idea. 'A' follows as it takes what is said in 'E' forward - it will be audible even if you don't want to listen to it. 'B' follows as it gives more details about connectivity during air travel'. D concludes the paragraph by providing an additional piece of information. Hence, CEABD is the right sequence.

Q40 QSPR

Solution

The sequence QSPR fits the context by first introducing the unique manner in which young Lincoln managed the shop (Q), followed by the description of his way of tending to a chance customer after reading (S). Then, a chance customer's arrival is mentioned (P), leading to Lincoln attending to their needs and returning to his reading (R). This sequence creates a logical flow by introducing Lincoln's atypical approach to managing the shop.

The correct answer is Option (2)

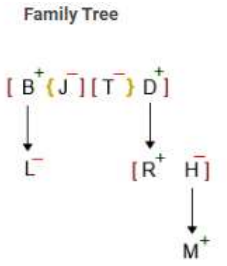
Q41 (A)

Solution

Substitute brethren for brethrens (brethren are plural which means brothers).

Q42. Aunt

Solution



Hence, the option A is correct.

Q43. Grandson

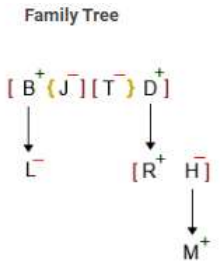
Solution

M is the son of H. T is the mother-in-law of H.

From this, we can say that M is the grandson of T

Q44. Niece

Solution



Hence, the option A is correct.

B, J, and T are siblings. B is the brother of T and L is the daughter of B.

So,L is the niece of T

Q45. 2

Solution

Statement (A) alone is not sufficient since it implies only that n = 2k + 3, which could be any odd number.

Statement (B) alone, is not sufficient since it gives only the k value and does not give any information about n.

After combining it is possible to find the value of k and n.

Q46. 3

Solution

To find the volume of cube, we need to find the side of the cube

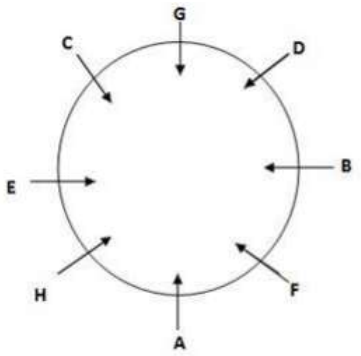
Statement (A) => side = face diagonal/√2 = √6/√2 = √3.

Statement (B) => side = body diagonal/√3 = 3/√3 = √3.

Both are independently enough.

Q47. C

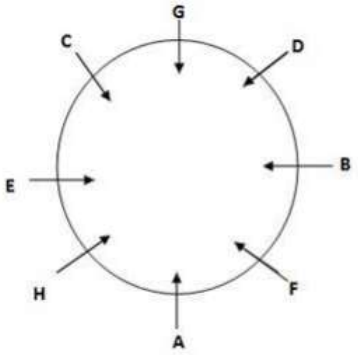
Solution



sits second to the left of H.

Q48. G

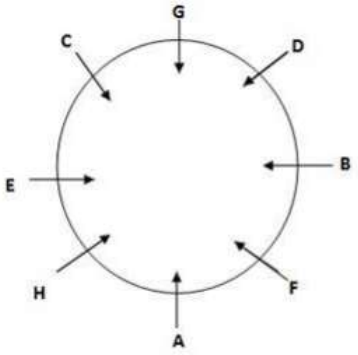
Solution



G sits second to the left of the person facing B.

Q49. A

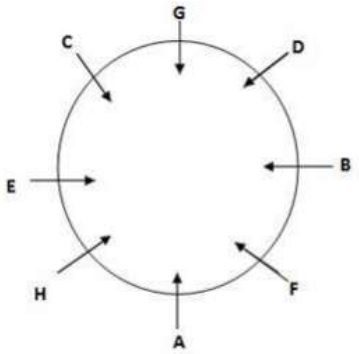
Solution



G is related to A.

Q50. None

Solution



o one will remain same.

Q51. it is light, making transportation easy and inexpensive.

Solution

The advantage of using fibre glass for sculptures is that it is light making transportation easy and inexpensive. (Refer to second paragraph last sentence). The right answer is "it is light, making transportation easy and inexpensive."

Q52. It is made of fibre-glass.

Solution

In the past, there were instances of non-resident Indian communities organising puja festivals abroad and placing orders for images of the Goddess from Kolkata. However, it is rare to choose fibre-glass for sculpting the image. (Refer to paragraph two, sentence one). So the right answer is "It is made of fibre-glass."

Q53. Middlesex.

Solution

An order for a new image next year is expected from Middlesex. (Refer to the last line of the passage). The right answer is "Middlesex."

Q54. did not immerse the idol at the close of the festival.

Solution

The organisers in Middlesex did not immerse the idol at the close of the festival. (Refer to para four second sentence). The right choice is "did not immerse the idol at the close of the festival."

Q55. an artist in the clay-modelling business.

Solution

Ir. Prodyut Pal is an artist in the clay-modelling business. (Refer to para three, sentence 2).

he right answer is "an artist in the clay-modelling business."

Q56. unique

Solution

In the context of the sentence, it mentions a "mobile network" offering a technology platform. The word "**unique**" is appropriate here because it indicates that the mobile network provides a distinctive and unparalleled technology platform onto which other services can be added.

Q57. undertake

Solution

The word "**undertake**" fits well in this context because it means to take on or commit to the task of carrying out financial transactions. When individuals use mobile banking, they are actively undertaking or performing various financial transactions through their mobile devices.

Q58. limited

Solution

"**Limited**" in this context suggests that mobile payment models are not yet extensively adopted or implemented on a large scale.

Q59. channels

Solution

"**channels**" refers to the communication methods or pathways through which services are delivered.The term "channels" here encompasses various communication methods that enable the delivery of mobile banking services, and SMS is mentioned as one example.

Q60. amenable

Solution

The term "**amenable**" means open and responsive to suggestion, willing to cooperate, or easily influenced. In the context of the sentence, the low cost of using existing

ıfrastructure makes mobile banking services more amenable to use by customers with

ıwer purchasing power.

Section 2 - Pseudocode and Computer Fundamentals

Q1

2

Solution

After evaluating the expression p becomes 8, so if block gets executed and prints 2 (i.e 1*2=2)

Q2

24

Solution

When c=2, updated value of a= 15,b=8.

When c=3 , b updated to 9 and else block gets executed and loop terminates.

So 15+9 is 24.

Q3

4

Solution

Initially a=1, b= 3, c= 2

If block condition gets failed,so 'b' value is set to 1. (b=3&1 => b=1)

As a result, 1+1+2=4

Q4

1

Solution

No Solution

Q5

37

Solution

Initially a=4,b=5,c=9, If block gets executed and update b=24.So 4+24+9=37.

Q6

32

Solution

Initially a=10, then updated to 11. If block gets executed and 'a' becomes 30. So 30+3-1=32.

Q7

57

Solution

Final value of a after executing first for loop is 11. Then inside second for loop,

when c=4,updated b value is 6.

when c=5,updated b value is 26.

when c=6,updated b value is 46.

So 11+46=57

Q8

31

Solution

a=4+2 = 6 , b=10&2 = 2 , b= (15&4)+6 =10. So 6+10+15 = 31

Q9

70

Solution

When c=4, updated value is a=19,b= 17.

When c=5, updated value is a=43,b= 29. Same process repeated until c becomes 8. So finally a=43,b=27. Therefore 43+27=70.

Q10

6

Solution

Relational Operators have higher precedence than arithmetic operators. So 'p' and 'q' values will get updated to 2.

As a result 2+2+2 = **6**

Q11

38

Solution

Initially a=9,b=9,c=8. If blocks gets executed and update b=9+11^8 = 28, c= 1. So 9+28+1=38.

Q12

48

Solution

Initially a=1,b=4,c=6, after performing the sequence of operation a=27 ,b=4 ,c=17. So 27+4+17 =>48

Q13

It will print cube of value of p

Solution

Second if block gets failed because p=24. So it will print cube of value of p.

Q14

12

Solution

Initially a=3,b=4,c=5 first if condition gets failed so 3+4+5=12 gets printed.

Q15

13

Solution

Inside for loop if block will not get executed for all the iterations. So flag value remains zero and print 13.

Q16

AND and NOT gates

Solution

No Solution

Q17

when each decimal digit is represented by four bit binary

Solution

No Solution

Q18

seventh layer

Solution

No Solution

Q19

Output device

Solution

No Solution

Q20

User data

Solution

No Solution

Q21

160

Solution

No Solution

Q22

Line Printer

Solution

The computer device primarily used to provide hard copies is the Line Printer.

A line printer prints one entire line of text before advancing to another line. Most early-line printers were impact printers.

Q23 Storage

Solution

No Solution

Q24 11000011

Solution

No Solution

Q25 2

Solution

No Solution

Q26 Super Computer

Solution

A supercomputer is a computer with a high level of performance compared to a general-purpose computer. The performance of a supercomputer is measured in floating-point operations per second (FLOPS) instead of million instructions per second (MIPS).

Q27 Satellite

Solution

No Solution

Q28 High Speed, Large Capacity Disk

Solution

No Solution

Q29 Joystick

Solution

No Solution

Q30 Bits

Solution

No Solution